Supplementary Material for "Integrated-Haar wavelet collocation with a provably real spectrum on arbitrary grids"

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The proofs are given in the appendices of the main text. Part B contains additional tables and a figure, all generated by the scripts of the repository https://github.com/mossaiby/iHaar.

Part B. Additional tables and figure

Table S1: Dependence on Poisson's ratio, prescribed displacements on all edges (smooth solution).

	GMRES iterations			e_{H^1}		
ν	$N = 32$	64	128	$N = 32$	64	128
0.3	17	17	16	5.78×10^{-4}	1.45×10^{-4}	3.61×10^{-5}
0.4	23	22	22	7.63×10^{-4}	1.91×10^{-4}	4.78×10^{-5}
0.45	30	30	29	1.17×10^{-3}	2.95×10^{-4}	7.39×10^{-5}
0.49	52	57	56	4.73×10^{-3}	1.23×10^{-3}	3.11×10^{-4}
0.499	96	124	141	3.14×10^{-2}	1.07×10^{-2}	2.96×10^{-3}

Table S2: Traction on two adjacent edges: dependence on Poisson's ratio.

	GMRES iterations			e_{H^1}		
ν	$N = 32$	64	128	$N = 32$	64	128
0.3	27	27	27	1.08×10^{-3}	2.70×10^{-4}	6.75×10^{-5}
0.4	33	33	33	1.93×10^{-3}	4.84×10^{-4}	1.21×10^{-4}
0.45	44	44	44	3.69×10^{-3}	9.29×10^{-4}	2.33×10^{-4}
0.49	102	98	98	1.75×10^{-2}	4.49×10^{-3}	1.13×10^{-3}
0.499	317	377	406	1.37×10^{-1}	4.15×10^{-2}	1.10×10^{-2}

Table S3: One-dimensional model $u'' = f$ with $u(0)$ and $u'(1)$ given: maximum error with an enhanced end row of weight α.

α	$N = 32$	64	128	256	512	rate
$-1/12$	4.0×10^{-4}	1.0×10^{-4}	2.6×10^{-5}	6.5×10^{-6}	1.6×10^{-6}	2.00
0 (plain)	1.2×10^{-3}	3.0×10^{-4}	7.6×10^{-5}	1.9×10^{-5}	4.8×10^{-6}	2.00
1/4	5.4×10^{-3}	1.4×10^{-3}	3.4×10^{-4}	8.6×10^{-5}	2.2×10^{-5}	2.00
1/2	9.5×10^{-3}	2.4×10^{-3}	6.1×10^{-4}	1.5×10^{-4}	3.8×10^{-5}	2.00

Table S4: Spectrum of D_2 on random graded grids: largest relative imaginary part of the eigenvalues and largest eigenvector condition number.

ends	ratio bound	N	grids	max $\lvert\mathrm{Im}\,\lambda\rvert/\lvert\lambda\rvert$	max $\kappa(V)$
values	1.5	16	1000	0.0e+00	3.1
values	1.5	64	400	0.0e+00	8.8
values	1.5	256	60	0.0e+00	20.2
values	2	16	1000	0.0e+00	4.3
values	2	64	400	0.0e+00	11.8
values	2	256	60	0.0e+00	31.4
values	4	16	1000	0.0e+00	6.4
values	4	64	400	0.0e+00	15.4
values	4	256	60	0.0e+00	43.0
zero slope	1.5	16	1000	0.0e+00	4.4
zero slope	1.5	64	400	0.0e+00	14.8
zero slope	1.5	256	60	0.0e+00	25.6
zero slope	2	16	1000	0.0e+00	7.2
zero slope	2	64	400	0.0e+00	22.2
zero slope	2	256	60	0.0e+00	54.8
zero slope	4	16	1000	0.0e+00	10.9
zero slope	4	64	400	0.0e+00	35.2
zero slope	4	256	60	0.0e+00	100.3

Table S5: Planar interface ($\varepsilon = 0.01$): maximum nodal error for three operators.

N	nodes per $\sqrt{2}\varepsilon$	3-point FD	D_2	$D_2^{(4)}$
48	0.68	1.47×10^{-1}	4.10×10^{-2}	9.77×10^{-2}
64	0.91	1.21×10^{-1}	1.29×10^{-2}	6.29×10^{-2}
96	1.36	4.07×10^{-2}	7.32×10^{-3}	1.24×10^{-2}
128	1.81	1.54×10^{-2}	6.45×10^{-3}	2.89×10^{-3}
192	2.72	8.02×10^{-3}	3.48×10^{-3}	4.88×10^{-4}
256	3.62	4.20×10^{-3}	1.99×10^{-3}	1.66×10^{-4}
384	5.43	1.90×10^{-3}	9.23×10^{-4}	3.16×10^{-5}
512	7.24	1.07×10^{-3}	5.28×10^{-4}	9.78×10^{-6}

Table S6: Cahn–Hilliard: temporal convergence of SAV-BDF2 ($N = 128$, $D_2^{(4)}$).

Δt	error	rate
1.60×10^{-4}	8.12×10^{-5}	–
8.00×10^{-5}	2.05×10^{-5}	1.99
4.00×10^{-5}	5.13×10^{-6}	1.99
2.00×10^{-5}	1.29×10^{-6}	2.00
1.00×10^{-5}	3.21×10^{-7}	2.00

Table S7: Spinodal decomposition: conservation, energy and cost.

quantity	2D spinodal	3D spinodal
grid	256^2	96^3
ε	0.01	0.02
time steps	5000+4500	1500
initial mass M_h^0	0.1	0.1
largest mass deviation $\lvert M_h^n - M_h^0\rvert$	9.71×10^{-17}	0
largest step change of the modified energy	-3.31×10^{-8}	-4.79×10^{-9}
largest change of E_h between samples	-8.47×10^{-7}	-2.43×10^{-7}
range of $\xi = r/\sqrt{E_1 + C}$	1.00000–1.00035	–
CPU time per step (one core)	1.9 ms	63 ms

Table S8: Maximum nodal error and observed convergence rates for two model problems.

(a) Prescribed values, $-\Delta u = f$, operator D_2

	uniform grid		Chebyshev-graded grid	
N	error	rate	error	rate
8	6.50×10^{-3}	–	3.96×10^{-3}	–
16	1.66×10^{-3}	1.97	1.15×10^{-3}	1.79
32	4.18×10^{-4}	1.99	2.96×10^{-4}	1.95
64	1.05×10^{-4}	2.00	7.46×10^{-5}	1.99
128	2.61×10^{-5}	2.00	1.87×10^{-5}	2.00
256	6.54×10^{-6}	2.00	4.67×10^{-6}	2.00
512	1.63×10^{-6}	2.00	1.17×10^{-6}	2.00

(b) Zero-slope conditions, $(I - \Delta)u = f$, uniform grid

	D_2		$D_2^{(4)}$	
N	error	rate	error	rate
8	3.05×10^{-2}	–	4.69×10^{-3}	–
16	9.11×10^{-3}	1.74	3.15×10^{-4}	3.89
32	2.38×10^{-3}	1.94	2.01×10^{-5}	3.98
64	6.01×10^{-4}	1.98	1.26×10^{-6}	3.99
128	1.51×10^{-4}	2.00	7.87×10^{-8}	4.00
256	3.77×10^{-5}	2.00	4.92×10^{-9}	4.00
512	9.42×10^{-6}	2.00	3.08×10^{-10}	4.00

Table S9: Smooth solution with prescribed displacements on all edges.

N	unknowns	e_{H^1}	rate	e_∞	GMRES its
8	128	9.22×10^{-3}	–	4.13×10^{-3}	15
16	512	2.31×10^{-3}	2.00	1.03×10^{-3}	17
32	2048	5.78×10^{-4}	2.00	2.55×10^{-4}	17
64	8192	1.45×10^{-4}	2.00	6.35×10^{-5}	17
128	32768	3.61×10^{-5}	2.00	1.58×10^{-5}	16
256	131072	9.04×10^{-6}	2.00	3.95×10^{-6}	15

Table S10: Traction boundaries, smooth solution: relative H^1 error for four configurations, GMRES iterations, and maximum displacement error on the traction edges.

	traction on T		traction on T+R	
N	e_{H^1}	rate	e_{H^1}	rate
16	3.30×10^{-3}	–	4.31×10^{-3}	–
32	8.20×10^{-4}	2.01	1.08×10^{-3}	2.00
64	2.05×10^{-4}	2.00	2.70×10^{-4}	2.00
128	5.11×10^{-5}	2.00	6.75×10^{-5}	2.00
256	1.28×10^{-5}	2.00	1.69×10^{-5}	2.00
GMRES its	18–20		27–27	
edge error, N=256	8.64×10^{-6}		1.84×10^{-5}	

	traction on L+B		traction on T+B	
N	e_{H^1}	rate	e_{H^1}	rate
16	3.89×10^{-3}	–	5.99×10^{-3}	–
32	9.70×10^{-4}	2.00	1.49×10^{-3}	2.01
64	2.42×10^{-4}	2.00	3.71×10^{-4}	2.00
128	6.05×10^{-5}	2.00	9.27×10^{-5}	2.00
256	1.51×10^{-5}	2.00	2.32×10^{-5}	2.00
GMRES its	26–27		18–20	
edge error, N=256	1.09×10^{-5}		1.53×10^{-5}	

Table S11: Traction on two adjacent edges, smooth solution, on graded grids.

(a) Chebyshev grid

N	e_{H^1}	rate	edge error	GMRES its
16	6.73×10^{-3}	–	8.10×10^{-3}	26
32	1.69×10^{-3}	1.99	2.04×10^{-3}	26
64	4.23×10^{-4}	2.00	5.10×10^{-4}	26
128	1.06×10^{-4}	2.00	1.27×10^{-4}	26

(b) Geometric grid graded toward the traction edge

N	e_{H^1}	rate	$h_{\max}/h_{\min}$	GMRES its
16	2.79×10^{-2}	–	66	39
32	7.55×10^{-3}	1.89	76	50
64	1.93×10^{-3}	1.97	82	69
128	4.87×10^{-4}	1.99	85	139

Table S12: Corner singularity with prescribed displacements: uniform versus automatic refinement.

(a) Uniform refinement

unknowns	e_{H^1}	GMRES its
128	9.62×10^{-2}	14
512	6.58×10^{-2}	16
2048	4.51×10^{-2}	17
8192	3.09×10^{-2}	16
32768	2.12×10^{-2}	16
131072	1.45×10^{-2}	15

(b) Automatic refinement (Haar-detail indicator)

cycle	unknowns	e_{H^1}	$h_{\min}$	$\kappa(V)$	GMRES its
0	128	9.62×10^{-2}	1.2×10^{-1}	1.0	14
2	220	6.46×10^{-2}	3.1×10^{-2}	2.0	15
4	312	4.45×10^{-2}	1.6×10^{-2}	2.5	15
6	480	3.20×10^{-2}	7.8×10^{-3}	2.9	15
8	756	1.66×10^{-2}	2.0×10^{-3}	3.2	14
10	1456	8.07×10^{-3}	4.9×10^{-4}	3.3	13
12	3040	3.93×10^{-3}	1.2×10^{-4}	3.8	13
14	6380	1.91×10^{-3}	3.1×10^{-5}	5.2	12
16	13608	9.07×10^{-4}	7.6×10^{-6}	5.3	11
18	29280	4.37×10^{-4}	1.9×10^{-6}	7.1	10
20	62300	2.09×10^{-4}	4.8×10^{-7}	7.1	10
22	130048	1.20×10^{-4}	1.2×10^{-7}	9.5	9

Table S13: Allen–Cahn shrinking circle ($\varepsilon = 0.02$, $\Delta t = 0.05$): fitted slope of $R^2(t)$, deviation from $-2\varepsilon^2$, and the largest step changes of the modified and the original energy.

grid	operator	$\mathrm{d}R^2/\mathrm{d}t$	deviation	max $\Delta\tilde{E}$	max ΔE_h
64^2	D_2	-8.138×10^{-4}	1.72%	-8.09×10^{-6}	-1.62×10^{-4}
64^2	$D_2^{(4)}$	-8.008×10^{-4}	0.11%	-7.91×10^{-6}	-1.58×10^{-4}
128^2	$D_2^{(4)}$	-8.028×10^{-4}	0.36%	-7.92×10^{-6}	-1.58×10^{-4}
256^2	$D_2^{(4)}$	-8.030×10^{-4}	0.37%	-7.92×10^{-6}	-1.58×10^{-4}

Table S14: Cahn–Hilliard: spatial self-convergence in the maximum norm.

	D_2		$D_2^{(4)}$	
N	error	rate	error	rate
18	7.95×10^{-4}	–	1.36×10^{-4}	–
54	9.32×10^{-5}	1.95	1.86×10^{-6}	3.91
162	9.40×10^{-6}	2.09	2.24×10^{-8}	4.02

Table S15: Uniform-grid operators within the identical SAV-BDF2 solver.

(a) Smooth Cahn–Hilliard, max error against a pseudo-spectral reference ($N = 486$)

scheme	$N = 18$	$N = 54$	$N = 162$
3-point FD	1.63×10^{-3}	1.89×10^{-4}	2.12×10^{-5}
IHWC, D_2	7.96×10^{-4}	9.44×10^{-5}	1.06×10^{-5}
IHWC, $D_2^{(4)}$	1.36×10^{-4}	1.86×10^{-6}	2.27×10^{-8}
compact (Padé)	9.42×10^{-5}	1.31×10^{-6}	1.60×10^{-8}
pseudo-spectral	4.40×10^{-7}	3.44×10^{-15}	1.61×10^{-15}

(b) Allen–Cahn star, $\varepsilon = 0.004$, against the IHWC $D_2^{(4)}$ reference (1024^2)

	max nodal error		area error	
scheme	256^2	512^2	256^2	512^2
3-point FD	3.89×10^{-1}	8.85×10^{-2}	6.02×10^{-4}	1.46×10^{-4}
IHWC, D_2	1.36×10^{-1}	4.12×10^{-2}	2.43×10^{-4}	6.88×10^{-5}
IHWC, $D_2^{(4)}$	5.27×10^{-2}	2.63×10^{-3}	6.07×10^{-5}	4.23×10^{-6}
compact (Padé)	3.65×10^{-2}	1.78×10^{-3}	3.77×10^{-5}	2.88×10^{-6}
pseudo-spectral	1.52×10^{-2}	1.68×10^{-4}	1.24×10^{-5}	2.72×10^{-7}

Table S16: Fast diagonalization (FDM) versus Bartels–Stewart (B–S) for $D_2U + UD_2^{\mathsf{T}} = F$ on the worst-conditioned of 20 random graded grids per row.

ratio	N	$h_{\max}/h_{\min}$	$\kappa(V)$	backward error FDM	backward error B–S	solution difference
1.5	32	1.1×10^{1}	3.0	1.4×10^{-16}	1.4×10^{-16}	3.5×10^{-15}
1.5	64	4.5×10^{1}	4.8	2.1×10^{-17}	1.9×10^{-17}	2.5×10^{-15}
1.5	128	5.9×10^{2}	10.2	1.4×10^{-17}	7.2×10^{-18}	7.8×10^{-15}
1.5	256	2.0×10^{3}	16.4	1.1×10^{-17}	6.8×10^{-18}	5.4×10^{-13}
2	32	8.3×10^{1}	5.5	3.2×10^{-17}	3.6×10^{-17}	9.9×10^{-16}
2	64	3.3×10^{2}	8.3	2.2×10^{-17}	2.3×10^{-17}	3.3×10^{-15}
2	128	7.5×10^{3}	16.6	4.0×10^{-17}	4.1×10^{-17}	1.8×10^{-15}
2	256	1.4×10^{4}	27.6	4.8×10^{-17}	2.3×10^{-17}	1.5×10^{-10}
4	32	2.5×10^{4}	10.2	9.8×10^{-18}	8.8×10^{-18}	2.5×10^{-13}
4	64	4.2×10^{3}	12.1	6.1×10^{-17}	9.0×10^{-17}	3.0×10^{-10}
4	128	1.1×10^{3}	20.0	2.1×10^{-16}	2.2×10^{-16}	8.0×10^{-10}
4	256	1.2×10^{6}	46.4	5.0×10^{-18}	5.4×10^{-18}	9.6×10^{-8}

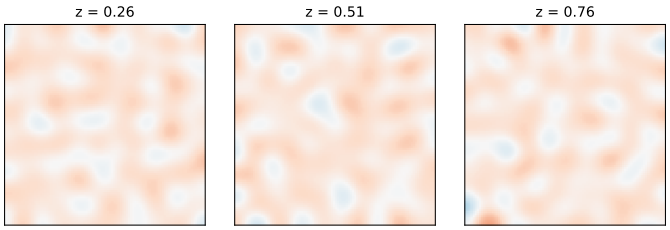


Figure S1: Three-dimensional Cahn–Hilliard on 96^3: slices of the concentration at the final time.